JomiCheck · কেনার আগে যাচাই করুন

Check before you buy: lawyer-grade pre-purchase land verification for citizens of Bangladesh

LandTech Innovation Hackathon 2026 · Concept Note
Team Semicolons
Track: Digital Land Records, Verification & Digitization
Problem: **P5** Citizen Pre-Purchase Verification, with **P3** Fraud & Double-Selling Detection

A buyer enters the mouza, dag number and seller's name, or uploads the papers the seller handed over. JomiCheck checks the land the way an experienced land lawyer would, and in about a minute returns a Low / Medium / High risk score in which **every finding names the law behind it, explains it in plain Bangla or English, and says what to do next**. The report is sealed with a SHA-256 hash and a QR code so a bank, lawyer or family member can confirm it was not edited.

1. The problem

For most families in Bangladesh, buying land is the largest purchase of their lives, and one of the riskiest. The facts that decide whether a sale is safe are spread across many records and offices: the khatian (record of rights) from four successive surveys (CS, SA, RS, BS), mutation (namjari) cases at the AC Land office, registered deeds at the sub-registry, court cases, government khas and acquisition lists, bank mortgages, powers of attorney, and the land tax (dakhila) office. An ordinary buyer cannot read and cross-check all of these, so they rely on the seller's word and on papers the seller provides. That gap is where fraud happens:

Risk	What happens to the buyer
Seller is not the owner	Someone sells land recorded in another person's name, or land of a deceased owner without the other heirs.
Double selling	The same dag, or the same share, was already sold by registered deed or promised by bainanama to another buyer.
Over-selling by heirs	An heir sells more than his Faraiz share; siblings and the widow later sue.
Forged papers	A fabricated khatian, a deed number that does not exist, or a tax receipt showing taxes paid that were never paid.
Restricted land	Khas, vested, waqf or forest land; acquisition notices; injunctions; mortgages.
Changes after checking	Between the bainanama and registration, the seller sells again, mortgages the land, or a case is filed.

The cost is lost savings, years in civil courts and broken family relationships. Citizens need a way to **screen a plot before paying any money**, in their own language, without needing to be a lawyer.

2. Our solution

JomiCheck is a working web application (Django REST API + Next.js, bilingual Bangla/English, mobile-friendly). It offers four entry points:

- **Verify land:** choose District → Upazila → Mouza, enter the dag number, the seller's name and the area being bought.
- **Upload a document:** upload a khatian or porcha photo or PDF; the AI reads it and fills the form.
- **Detect forged papers:** upload up to five papers (khatian, deed, tax receipt); each one is compared with the official record.
- **Check a report:** scan a report's QR code to confirm it is genuine and unchanged.

Land details or papers → Collect every record of the dag → Identify the seller's right → 25 legal checks → Faraiz shares → ML anomaly model → Risk score → Sealed report → Ask & watch

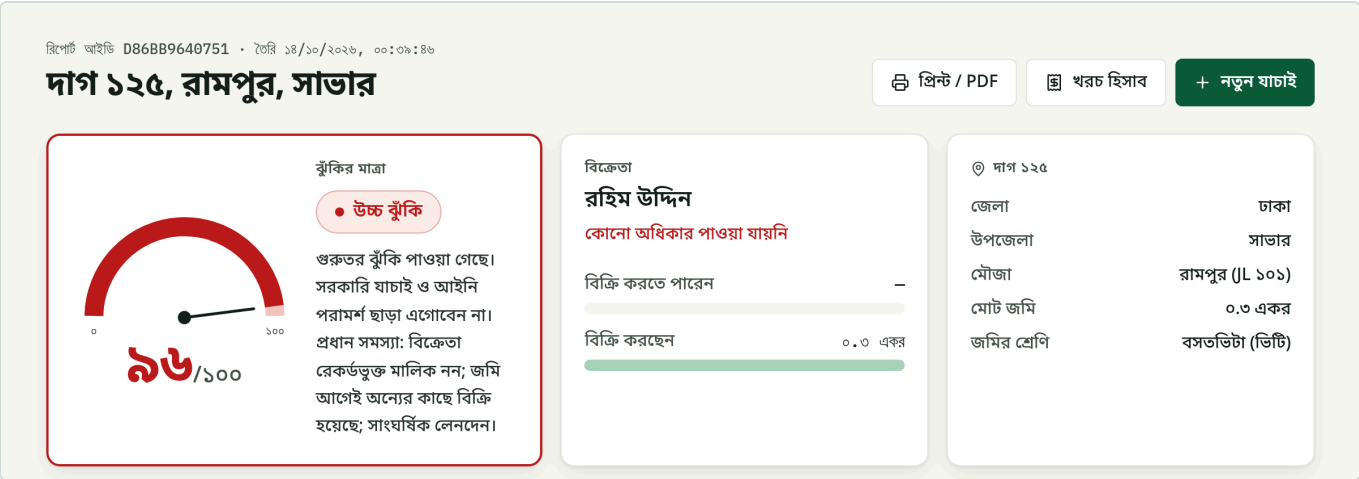


Figure 1. Report for the planted fraud case (dag 125, Rampur): the seller has no recorded right and the land was already sold, so the result is High risk. The full report continues with each finding, its law and the action to take.

3. How it works

3.1 Seller identification

The seller's name, in Bangla or English, is matched against the recorded owners of the latest khatian, the heirs of deceased owners, buyers in later registered deeds, and holders of a power of attorney. Matching uses RapidFuzz with Bangla–English transliteration and prefix handling (e.g. "Md." / "মোঃ"), so "Md. Rahim Uddin" and "মোঃ রহিম উদ্দিন" are recognised as the same person. The result decides which rules apply: an owner, an heir, a deed buyer and an attorney each have different legal requirements.

3.2 Legal rule engine (25 checks)

Each rule is written once in a catalog with its tier, weight, bilingual explanation, recommended action, a question to ask the seller, and the law it comes from (Registration Act 1908, Transfer of Property Act 1882, State Acquisition and Tenancy Act 1950, Land Crime Prevention and Remedy Act 2023, Land Development Tax Act 2023, Power of Attorney Act 2012, Acquisition Act 2017, Guardians and Wards Act 1890, CPC Order XXXIX, Muslim personal law).

Tier	Examples	Effect
Critical	Seller not the owner; recorded owner deceased; land already sold; khas, vested, waqf or forest land; acquisition notice; injunction; invalid power of attorney; uploaded paper contradicts the record	+35 to +40; result is High regardless of score
Major	Pending case; break in the survey chain; heir over-selling his Faraiz share; minor's land sold without court permission; mortgage; rapid transfers; mutation missing; prior bainanama; ML-detected unusual transfer	+20 to +30
Minor	Conflicting deeds; pre-emption (Hak-e-Shufa) risk; wrong khatian number; under-priced past deed; tax arrears; sale through a valid attorney; other buyers checked this dag	+5 to +10

The score is capped at 100: 0–30 Low, 31–60 Medium, 61–100 High. Because every point comes from a named rule, the result is never a black box.

3.3 Inheritance (Faraiz) calculator

When the recorded owner has died, JomiCheck computes each heir's share under Muslim personal law (spouse, sons, daughters, father, mother, including 'awl, radd and the Umariyyatan cases) and checks whether the heir selling is offering more than his share. For non-Muslim owners it asks for a lawyer's review rather than guessing.

3.4 Forged paper detection

Gemma 4 (Google AI Studio) reads each uploaded paper into structured fields. **The comparison itself is done by code, not by the AI**: does this khatian number exist in this mouza; do owners and shares (in ana-gonda-kora-kranti or fractions) match; are the dags and area right; does the deed number exist in the sub-registry record, or was it cancelled; are the sellers, buyers and date correct; does the tax receipt's holding number and paid-up year match the tax record. A paper that contradicts the record raises a Critical finding under the Land Crime Prevention and Remedy Act 2023 (s.5) and the Penal Code (ss.467, 471).

3.5 Machine learning: unusual transfer detection

An Isolation Forest (scikit-learn) scores every sale of the plot in the last ten years plus the proposed sale on five features: price ÷ government mouza rate, how long the seller held the land, sales of the dag within a year, share of the dag sold, and dags sold by the same seller within a year. It catches combinations no single rule catches, such as a broker quickly reselling many dags. Trained on 20,000 synthetic transfers with 5% planted fraud patterns, it reaches on a separate synthetic test set of 5,000:

Precision	Recall	F1	False-positive rate
0.745	0.712	0.728	0.013

These numbers show that the pipeline works on synthetic data; they are not real-world accuracy. The model never decides alone that land is unsafe: it adds evidence to a rule or raises one Major finding with plain-language reasons.

3.6 The report

- Risk gauge and a plain-language summary; each finding with what was found, the law, the action and a question for the seller.
- The latest khatian in the official column layout, the survey lineage (CS → SA → RS → BS → mutation), the full ownership timeline and the heirs' Faraiz table.
- A **mouza map** with the buyer's dag highlighted and neighbouring dags coloured by risk, so boundary disputes can be anticipated.
- Document checklist, site-visit checklist, purchase cost estimate (stamp duty, registration fee, taxes, mutation fees) and a print/PDF version.
- A SHA-256 seal and QR code; the "Check a report" page shows whether a copy matches the original.
- **Ask about this report:** Gemma answers questions in Bangla or English using only the report's facts; it explains the score but never sets it.
- **Plot watch:** the buyer can watch the plot until registration. Any deed, mutation, case, mortgage, bainanama or power of attorney recorded later, or another buyer checking the same dag, appears as an alert.

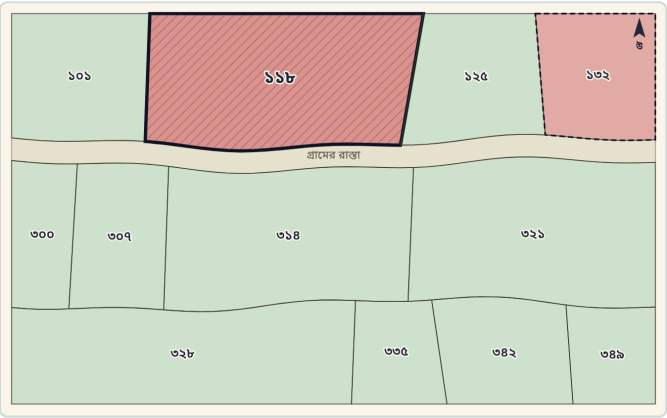


Figure 2. Mouza map: the buyer's dag (118) is outlined; dag 132 next to it is High risk because it was already sold. Outlines are generated, not surveyed.

3.7 Demonstration data

14 planted scenarios cover the main fraud types, from a clean sale (Low) to double selling, khas land, an injunction, a revoked power of attorney, a widow selling a minor's share, and a broker pattern that only the ML model catches. Three sets of sample papers include a forged khatian, a deed whose number is not in the sub-registry record, and a tax receipt showing taxes paid two years beyond the record.

The prototype is deployed on AWS EC2 and covered by 71 automated tests (Faraiz, every demo scenario, khatian notation, costs, ML model, document cross-check, plot watch, map, API and tamper detection).

4. Technology

Layer	Choice and reason
Backend	Python 3, Django 6 + Django REST Framework; SQLite in the prototype, PostgreSQL-ready. Rules, Faraiz and scoring are plain, tested Python.
Frontend	Next.js 16, React 19, Tailwind CSS; full Bangla/English interface; works on a phone.
AI	Gemma 4 (open-weight model, via Google AI Studio) for reading documents and answering questions. An open model can later run on government servers so citizen documents never leave the country.
ML and matching	scikit-learn Isolation Forest; RapidFuzz for Bangla/English name matching.
Integrity, hosting	SHA-256 hash, QR code and public check page; AWS EC2 with nginx, gunicorn and systemd.

5. Who benefits and how it can be adopted

- **Buyers**, especially first-time and rural buyers and migrant workers buying from abroad: a clear answer before paying the bainanama money.
- **Banks and NGOs** lending against land: a sealed, verifiable pre-check of the collateral.
- **Lawyers, deed writers and sub-registry offices:** a checklist of what to verify, which shortens due diligence.
- **Government:** fewer land disputes reaching the courts, and early signals of forged papers and broker networks.

Path to real use: the engine already works on the same structure as the national records (mouza, JL, dag, khatian of each survey, mutation case, deed). Connecting it to authorised APIs of e-Porcha, e-Mutation, the land development tax system and sub-registry deed records replaces the synthetic database without changing the rules. Our first target is a pilot in one upazila with the AC Land office, offered free to citizens; banks and institutions could pay for bulk or API checks.

6. Data, ethics and limitations

- **All data is synthetic.** Every person, plot, khatian, deed, case and NID is fictional; only district and upazila names are real. No government portal is accessed or scraped.
- JomiCheck is **decision support**, not official verification or legal advice. Every report says so and directs the buyer to the AC Land office, sub-registry and a lawyer.
- Uploaded documents are processed in memory and never stored. The AI never decides the risk; rules do, and every rule is visible.
- The ML accuracy is measured on synthetic data and must be re-trained on real, authorised data. Mouza map outlines are generated; real deployment needs digital cadastral maps from the Directorate of Land Records and Surveys.
- Automatic inheritance shares currently cover Muslim law only. Legal content is being reviewed with our faculty mentor and the Department of Law.

7. Plan for the final round

1. Legal review of every rule and its citation with a practising land lawyer and our faculty mentor.
2. Hindu (Dayabhaga) and Christian inheritance rules; more document types (DCR, mutation order, heirship certificate).
3. User testing with ten buyers and two deed writers; simpler language where they struggle.
4. SMS alerts for plot watch, HTTPS with a domain, and a printable Bangla report tuned for banks.
5. Run Gemma on our own server for privacy, and a fallback reader when the cloud model is unavailable.

8. AI tools used (disclosure)

Cursor (AI coding assistant) helped scaffold the Django and Next.js projects, draft code, write tests and documentation; all code was reviewed, run and tested by the team. **Gemma 4 (Google AI Studio)** is part of the product: it reads uploaded documents and answers questions about a report, using synthetic data only.

Source code: github.com/nomancsediu/JomiCheck · Live prototype: http://18.143.183.93 · Team Semicolons: **[Team leader name]**, **[Member name]**, faculty mentor **[Mentor name]**, **[University]**